

LESSON 5.7

SCALE DRAWINGS



LESSON INTRODUCTION

MA.7.GR.1.5 Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

LEARNING TARGETS

- I can draw a scaled figure on grid paper given a scale factor.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles.

ESSENTIAL QUESTIONS

- How do architects and designers use scale drawings in their industries?
- How do you create a scale drawing using a scale factor?

LESSON NARRATIVE

Students integrate the concepts from Lessons 5 and 6 to construct their own scale drawings. Students will work in small groups to create a scale drawing of either an area of the school or one of the provided buildings. Each group will be assigned a landmark or area to draw and a scale factor to use. Students then compare and reflect on the differences between their scale drawings with a partner who completed a different one before practicing applying the concepts on their own.

COMPONENT SUMMARY

5.7.1 WARM-UP (5 MIN.)

Career Profiles – architectural drafter

5.7.3 COLLABORATION (10 MIN.)

Discuss scaled drawings with a partner from a different group

5.7.5 WRAP-UP (5 MIN.)

Self-assessment of understanding and interest in the day's learning target

5.7.2 ACTIVITY (15-20 MIN.)

Work with a real-world building or a room in the school to create a scale model

5.7.4 YOUR TURN! (10 MIN.)

Practice drawing scale drawings, determining scale factor, and using it to solve problems

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- Career Profile video
- Tools for measuring

ADDITIONAL PREPARATION

- Consider assigning group members ahead of time.
- Assign each group to a room or building.
- Assign each group a scale factor.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle to determine the scales used on their scale drawings. The scale should be able to be used to determine the actual measurements of the original landmark or room.

THINGS TO CONSIDER

- Consider formative assessment data collected up to this point in the unit when determining groupings and assignments for the 5.7.2 Activity.
- Communicate to students what decimal place they should round to when completing their scale drawings, if needed. Consider using the thousandths place.
- Students should be allowed to use a calculator during the activity to ensure focus on the drawing.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

5.7.3 Collaboration can be structured as a Think-Pair-Share to allow students the opportunity to reflect first before sharing with their new partner. The questions are designed to spark thinking for the individual learner. Allow each student to make a conjecture about the answers to the questions and share with a partner to explain their reasoning.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.6.1 Reasonableness

Multiple components throughout the lesson allow students to make connections to real-world experiences, including reflecting on how scale drawings are used in professional industries and making their own scale drawings of real-world landmarks or spaces.

DIFFERENTIATE INSTRUCTION

When creating the tables to display the original measurements and the measurements of the scaled drawings, have students draw arrows from the original to the scaled measurements and label each arrow with the scale factor they worked with.

Consider how to use the scale factors and assigned landmarks or spaces in the school as differentiation in assigning the 5.7.2 Activity.

Additional differentiation opportunities are denoted throughout the lesson guide.

Room/area/landmark being measured: <i>The Singing Tower</i>						
Scale factor: $\frac{1}{5}$						
	Side 1	Side 2	Side 3	Side 4	Side 5	Side 6
Actual Measurement	11 m	62 m	16 m			
Scale Drawing Measurement	2.2 m	12.4 m	3.2 m			

5.7.1 WARM-UP: CAREERS – DRAFTERS

TEACHER MOVES

Ask students if they can think of other fields or situations in which using scaled models might be important.

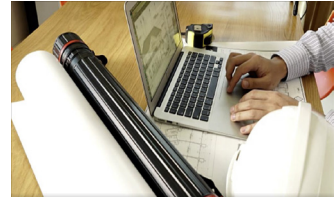
QUESTIONING

Do you know any drafters or people who work in architecture? Consider partnering with a local architecture firm to have a classroom visitor share how they use scale factor in their job.



5.7.1 Warm-Up: Careers - Drafters

Many jobs use mathematics. One such job is that of an architectural drafter.



Architectural drafters work in many industries to create drafts, or scale drawings, for different projects. They use specialized computer programs to create model drawings of buildings, wiring systems, topographical maps, bridges, and the electrical systems in many products. While many spend most of their time working on the computer, some architectural drafters do get to visit job sites. Training to become an architectural drafter requires either a specialized certificate or an associate degree.

1. What are some things that could go wrong on these projects if the scales are incorrect?

Answers will vary. Possible responses:

1. *If the scales are incorrect, there could be issues with wiring systems.*
2. *People could make costly mistakes such as ordering too much or too little of needed materials.*

2. Why is the use of scale drawings such a key part in so many industries?

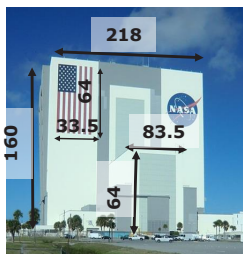
Answers will vary. Possible responses:

1. *Scale drawings are used in industries because the drawings would be too big to fit on paper if they were drawn at full size.*
2. *Scale drawings allow people to determine whether something will work before building it.*



5.7.2 Activity: Creating a Scale Drawing

Working with your small group, create a scale drawing of a specific room, area, or building. Each group will be assigned a different area as well as a different scale factor. If the rooms are not available, your group may choose from the landmarks below.

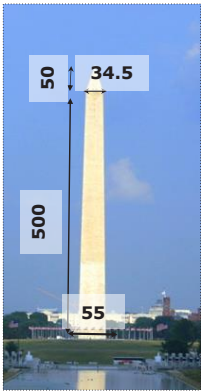
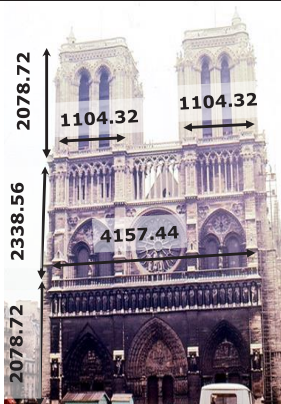
The Singing Tower, Bok Gardens Lake Wales, FL Measurements are in meters (m)	
NASA's Vehicle Assembly Building Kennedy Space Center in FL Measurements are in meters (m)	

5.7.2 ACTIVITY: CREATING A SCALE DRAWING

TEACHER MOVES

Assign one of the following scale factors to each group:

$$\frac{1}{5}, \frac{3}{8}, \frac{2}{5}, \frac{3}{4}, \frac{3}{5}, \frac{1}{10}, \frac{1}{9}, \frac{1}{8}, \frac{2}{9}, \frac{2}{7}$$

The Washington Monument Washington, D.C. Measurements are in feet (ft)	
Notre Dame Cathedral Paris, France Measurements are in centimeters (cm)	

DIFFERENTIATE INSTRUCTION

Consider students' comfort with the content and performing calculations when assigning landmarks and scale factors. Each student should be challenged at a level appropriate to their current understanding that will also help to push their understanding deeper. Consider partnerships in this framework as well.

5.7.2 ACTIVITY: CREATING A SCALE DRAWING (CONTINUED)

DIFFERENTIATE INSTRUCTION

Encourage students to draw arrows from the actual measurement to the scale drawing measurement to show the scale factor they multiplied by to find the measurement in the scale drawing. This provides an additional visual aid to students as they work through the table.

- Complete the table based on your assignment. Sides five and six are provided if a room has alcoves or closets. If columns are not needed, leave them blank.

Room/area/landmark being measured: <i>The Singing Tower</i>						
Scale factor: $\frac{1}{5}$						
	Side 1	Side 2	Side 3	Side 4	Side 5	Side 6
Actual Measurement	11 m	62 m	16 m			
Scale Drawing Measurement	2.2 m	12.4 m	3.2 m			

ENRICHMENT

As students work, circulate and ask guiding questions as needed, such as:

- What scale is appropriate given the number of gridlines available?
- Which dimensions are available for me to make my drawing?
- Because we're creating a scale drawing of a building that's already been created, some of our measurements are more approximations than exact. Why would an architectural drafter need to have the exact measurements?

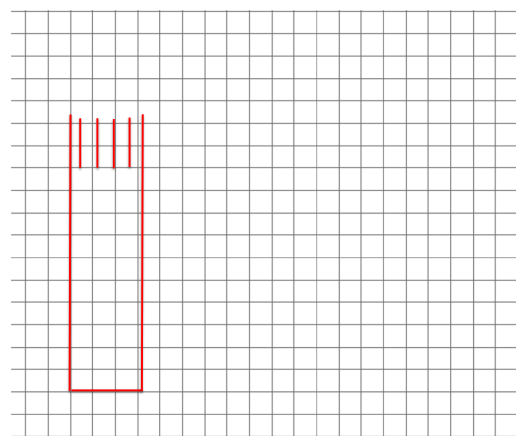
TEACHER MOVES

Students will be sharing their work with other groups in the next component. Encourage groups to discuss what they are doing to ensure all students are prepared to share and explain in the next component. If desired, have students preview the questions they will be reflecting on prior to moving to the next component.

- Create your scale drawing. Complete the title and scale for your drawing. For example, one gridline might equal two meters.

Scale Drawing of *The Singing Tower*

Scale: one gridline = 5 m





5.7.3 Collaboration: Numbered for Sharing

1. Number each member in your group. Partner with someone from a different group who has your same number. Discuss and record your answers to the following questions.

Questions	Reflections
Each of the scale factors assigned were fractions. Why do you think this makes sense for this assignment?	<i>Answers will vary. I think that fractions made sense because if they had been greater than 1, the scale drawing would have been too big.</i>
How did your scale differ from your partner's scale?	<i>Answers will vary. I said 1 gridline is equal to 5 meters while my partner drew their drawing with each gridline being 6 feet.</i>
How did the process of creating the scale model differ between you and your partner?	<i>Answers will vary. I found each side at a time, and my partner found them all at once before drawing.</i>
How did the different scale factors affect the final drawing?	<i>Answers will vary. The scale factors with larger denominators resulted in smaller drawings.</i>
Some groups may have turned their book sideways to create their scale drawing. Why do you think someone may have done that?	<i>Answers will vary. They may not have had enough room to draw the height to scale when their page was vertical.</i>

5.7.3 COLLABORATION: NUMBERED FOR SHARING

ENRICHMENT

Encourage students to consider these additional discussion questions if time allows:

- What do you notice about the smaller the scale factor?
- Why are we only working with scale factors that are less than zero?
- What would happen if a group was assigned to use a scale factor of 2 or 3?

5.7.4 YOUR TURN!

ENRICHMENT

Students may struggle to create a scaled copy of the figure with a perimeter of 27. Ask questions to further scaffold this section for students:

- What is the perimeter of the given figure? (10 cm; ensure students are counting each side of the L-shaped figure.)
- What relationship does that have to a perimeter of 27? (It is smaller than 27, so I will need to multiply by a whole or mixed number to create a figure with a perimeter of 27.)
- What can we multiply the length of each side by in order to obtain the new figure? (2.7.)

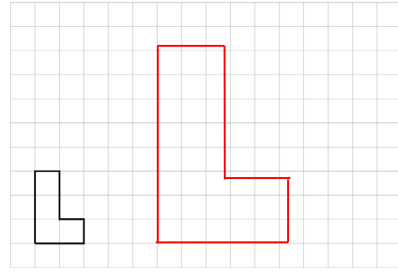
TEACHER MOVES

If students struggle responding to this prompt, have them rewrite the facts as $3\text{ m} = 6\text{ cm}$ and determine a scale factor from that. Then use that scale factor to calculate the cm for 27m. It may also be helpful to encourage students to draw arrows from 3m to 6cm and write the scale factor that results in the change. Then, students have a visual to determine the cm that represent 27m.



5.7.4 Your Turn!

1. A polygon is shown. Each gridline represents one centimeter (cm).



- a. Draw a scaled copy of this polygon that has a perimeter of 27 cm.
- b. Explain how to determine the scale factor.

Answers will vary. The perimeter of the scaled copy is equal to the perimeter of the original polygon times the scale factor i.e. $27 = 10 \times \text{scaled factor}$. Working backwards, we can determine that the scale factor is 2.7.

2. A scale on a blueprint drawing of a house shows that 6 cm represents 3 m. What number of centimeters on the blueprint represents an actual distance of 27 m?

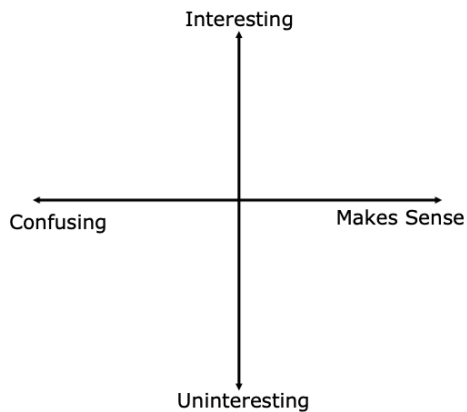
An actual distance of 3 m is equal to 6 cm on the blueprint. Therefore, an actual distance of 27 m is equal to 54 cm on the blueprint. $3 \times 2 = 6$ and $27 \times 2 = 54$

3. A scale on a hiking map shows that 3 inches (in.) represents 1.25 miles (mi.). What number of inches on the map represents 10 miles?

$1.25 \times 2.4 = 3$, and $10 \times 2.4 = 24$. 24 inches on the map would represent 10 miles.

**5.7.5 Wrap-Up: Reflection**

1. Plot a point in one of the quadrants below to indicate your understanding of today's learning target: "I can draw a scaled figure on grid paper given a scale factor."



Answers will vary.

5.7.5 WRAP-UP: REFLECTION**QUESTIONING**

For students who finish early, ask them to reflect on their answers and possibly answer one of the following questions:

- What do you think would make this lesson more interesting?
- What would make this lesson less confusing?
- If you could make any changes to the lesson, what would the changes be?